E-commerce Discount Framing A/B Test

₹200 OFF vs 20% OFF · Does the way a discount is shown change purchase conversion? · Synthetic portfolio project

The business problem

A ₹1,000 cart gets a ₹200 discount. The business can show it as "₹200 OFF" or as "20% OFF". The saving is exactly the same, but customers may react differently to how it is presented.

The question

Does the framing change how many people complete a purchase, and which one should the business prefer?

Experiment details

- 20,000 synthetic users, split equally: 10,000 per group
- Group A (Control): ₹200 OFF
- Group B (Treatment): 20% OFF
- Cart value is ₹1,000 for every user, so the saving is ₹200 in both offers

How success is measured

- Primary metric: purchase conversion rate
- Guardrails: complaint rate and return rate (among purchasers)
- Diagnostic: average order value (AOV)

Tools used: SQL · Python · Power BI

Explore the results



Same saving. Different framing.

₹200 OFF VS **20% OFF**

Control (Group A) | Treatment (Group B)

Note: Synthetic dataset created for portfolio practice. Not from a real company or a real experiment.

Which discount framing should the business prefer? ₹200 OFF vs 20% OFF

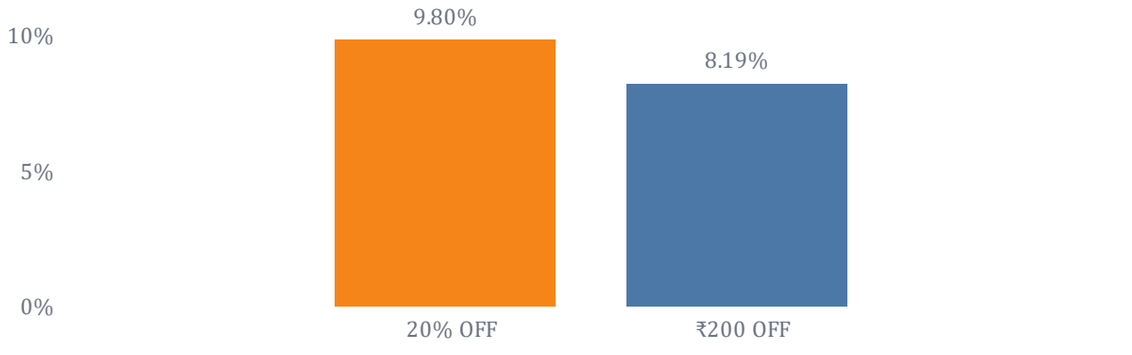
A/B test on 20,000 synthetic users · Primary metric: purchase conversion · Same ₹200 saving on a ₹1,000 cart

8.19%	9.80%	1.61	19.66%
Control: ₹200 OFF	Treatment: 20% OFF	Absolute Lift (pp)	Relative Lift

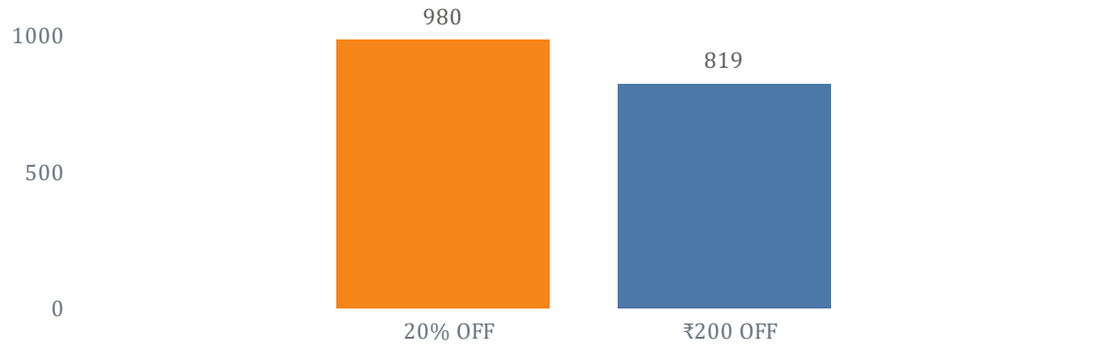
20000	3.98	p < 0.001	[0.82, 2.40] pp
Total Users	Z-statistic	Significance (α = 0.05)	95% CI of difference

20% OFF shows significantly higher conversion than ₹200 OFF (synthetic data)

Purchase Conversion Rate by Offer



Purchasers by Offer (out of 10,000 users each)

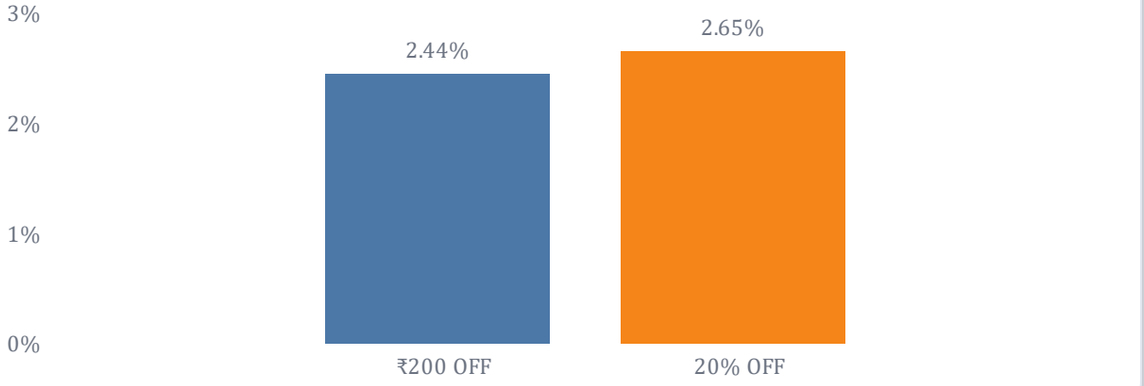


Note: Synthetic dataset created for portfolio practice; not from a real company or experiment. Test: two-proportion z-test (two-sided, α = 0.05). pp = percentage points

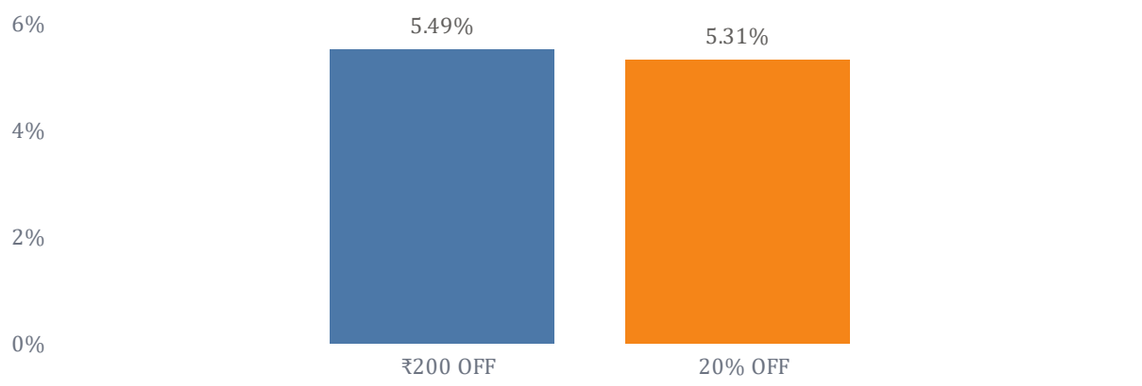
Did 20% OFF hurt customer experience? Guardrail checks

Complaint and return rates are calculated among purchasers only (819 in Control, 980 in Treatment)

Complaint Rate by Offer (among purchasers)



Return Rate by Offer (among purchasers)



0.21	p = 0.778	[-1.25, 1.67] pp
Difference (B – A, pp)	Significance	95% CI

-0.19	p = 0.860	[-2.29, 1.91] pp
Difference (B – A, pp)	Significance	95% CI

No statistical evidence that 20% OFF worsened complaints or returns (counts are small, so modest changes cannot be ruled out)

₹800	₹800	Identical by design
AOV: Control (₹200 OFF)	AOV: Treatment (20% OFF)	AOV sanity check

Note: Synthetic dataset for portfolio practice. Two-proportion z-tests, two-sided, α = 0.05. "Not significant" means no evidence of a change was found, not that the rates are identical. pp = percentage points.

Business decision: which discount framing should we prefer?

Summary of the evidence from this experiment (synthetic data, ₹1,000 cart)

Recommendation: consider 20% OFF framing, and confirm it with a real-world experiment before any rollout

+1.61 pp (19.66% relative lift)
Conversion result

Significant: p < 0.001, 95% CI [0.82, 2.40] pp
Statistical evidence

~161 (range ~82 to ~240)
Estimated extra purchases per 10,000 users

~₹128,800 (range ~₹65,381 to ~₹192,219)
Estimated extra order value per 10,000 users (not profit)

No evidence of harm: complaints p = 0.778, returns p = 0.860
Guardrail status

Decision scorecard

Decision check	Result	Status
Conversion improved (B vs A)	+1.61 pp	Pass
AOV unchanged	Identical by design	Pass
Statistically significant (p < 0.05)	p < 0.001	Pass
Complaint rate: no evidence of increase	p = 0.778	Pass
Return rate: no evidence of increase	p = 0.860	Pass

Note: Synthetic dataset for portfolio practice. Estimates are illustrative, not forecasts. pp = percentage points.